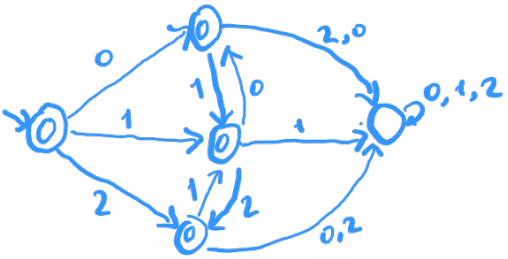
November 14, 2025 Midterm Duration: 90 minutes

Name: Solutions Corrections & Clarifications Extra Explanations Student No:

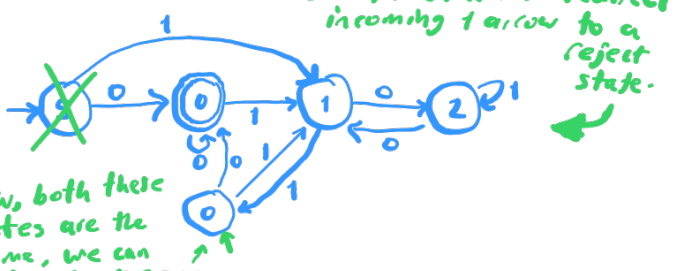
P1 [20 points] Construct a DFA for the languages

(a) $\{w \mid w \in \{0,1,2\}^*$ and $|w_i - w_{i+1}| = 1$ i.e. the difference between any consecutive digits in w is 1.$\}$ Use at most 5 states.



(b) $\{w \mid w \in \{0,1\}^*$ where w is a binary number divisible by 6. $\}$ Use at most 5 states to get 8 points, at most 4 states to get full 10 points.

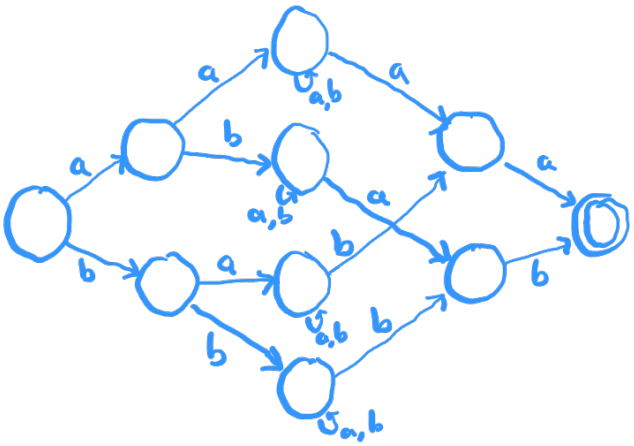
idea: must be div. by 3 and must end with 0. So we draw for div by 3 and split the accept state to redirect incoming 1 arrow to a reject state.



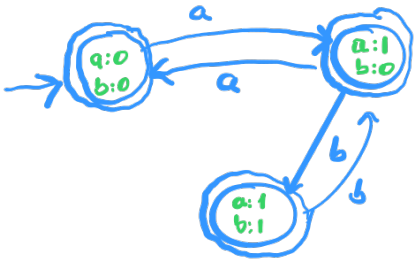
Now, both these states are the same, we can just make this one the start state and get rid of the other

P2 [20 points] Construct an NFA for the languages below. Use as few states as you can. Fewest gets full score, every extra state halves your score.

(a) $L = \{w \mid w \in \{a,b\}^*$, $|w| >= 4$, and w starts and ends with the same 2-letter substring.$\}$ For example $abbbbab \in L$ because it starts and ends with ab. Use at most 10 states.



(b) $L = \{w \mid w \in \{a,b\}^*$ and for any prefix x of w, $(n_a(x) \geq n_b(x)) \mod 2\}$ Use at most 3 states. (Number of as are never less than number of bs modulo 2 as we iterate through w. For example $abba \in L$. However $abab \notin L$ because for the prefix aba, $(n_a(aba) \geq n_b(aba)) \mod 2\}$ isn't satisfied.)



P3 [15 points] Prove that the language $\{1^k w \mid$ where $|w| \leq 2k$. $\}$ is not regular. ($w \in \{0,1\}^*$)

Assume that L is regular. Then, by PL for RL, $\exists p \in \mathbb{Z}^*$ s.t. $\forall s \in L$ with $|s| \geq p$, s can be splitted into xyz satisfying: 1. $\forall i \in \mathbb{N}\ xy^i z \in L$ 2. $|y| > 0$ 3. $|xy| \leq p$

Now, let $s = 1^p 0^{2p}$. To satisfy 2 & 3 we need $y = 1^m$ $m \geq 1$. But when we pump down, s' becomes $1^{p-m} 0^{2p} \notin L$ since $2p-m < 2p$.

So, our assumption must be wrong, L is not regular. QED.

P4 [30 points] Give the CFG and PDA for the following language: $\{w0^k1^k w^R \mid w \in \{0,1\}^*, k \geq 1\}$ where w^R means reverse of w, which means w written backwards.

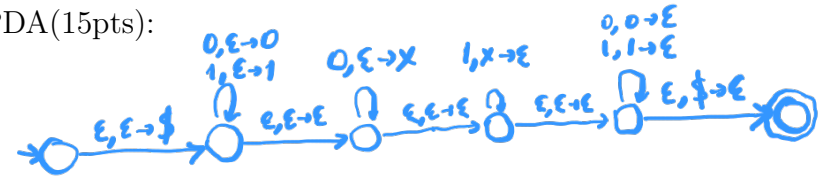
CFG(15pts):

$S \to 0S0 \mid 1S1 \mid A$

$A \to 0A1 \mid 01$

you cannot end with ε since $k \geq 1$.

PDA(15pts):



P5 [15 points] Prove that the following language is not context-free:

$$L = \{0^k 1^m 0^n \mid k, m, n \geq 0, 4k \leq 2m \leq n\}$$

Assume that L is CF. Then, by PL for CFL, $\exists p \in \mathbb{Z}^+$ s.t. $\forall s \in L$ with $|s| \geq p$, s can be splitted into uvxyz st. 1. $uv^i xy^i z \in L \; \forall i \in \mathbb{N}$
2. $|vy| > 0$
3. $|vxy| \leq p$.

Now, let $s = 0^p 1^{2p} 0^{4p}$. To satisfy cond. 1, we need to distribute v and y parts proportionally to 0s and 1s in all 3 sequences (0^p 1^{2p} and 0^{4p} parts) but this is impossible due to cond 3. So, our assumption was wrong, L is not CF. QED.